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MTBERT-Attention: An Explainable BERT Model based on Multi-Task Learning for Cognitive Text Classification

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A B S T R A C T

In recent years, there has been a lot of focus on Bloom's taxonomy-based classification of E-Learning materials. Researchers employ different methods and features. In our previous works, we have dealt with this problem via different techniques and algorithms. we started by boosting traditional machine learning algorithms then we moved on to deep learning algorithms by proposing a bidirectional GRU (BI-GRU) model combined with word2vec, finally we also proposed a fine-tuned model of bidirectional encoder representations from Transformers (BERT). The limitations of our model's performance as well as the lack of an annotated dataset lead us to explore a novel approach to the cognitive classification of text. First, we propose MTBERT-Attention, a unique and explainable model based on multi-task learning (MTL), BERT, and the co-attention mechanism. MTL enhances our primary task's generalization capacity and permits data augmentation. The usage of BERT as a stack for knowledge transfer between activities has been improved. The co-attention mechanism put particular emphasis on the significant aspects of the learning objective. We propose an explainability framework based on the attention mechanism. Lastly, we carry out in-depth tests to assess the viability and efficiency of the suggested model and the explainability Framework. Our proposed model outperforms the baseline models for loss, F1-score, and accuracy. It attained an overall classification accuracy of 97.71% with the test set and effectively classifies learning objectives that use unclear action verbs from Bloom's taxonomy. On the other hand, to prove the performance of our explainability Framework, we conduct a qualitative and quantitative study on the quality of the explanations as well as on the computational cost. We adopt Local Interpretable Model-agnostic Explanations (LIME) as a baseline for comparison. Our experiments show that our proposed approach for explainability outperforms the LIME explainer in terms of fidelity and computational cost.

Introduction

Reflecting on the learning objectives is the major goal of teaching and learning. However, teachers only have a few instruments at their disposal to help them [1]. In this way, the goal of our research is to help teachers by suggesting relevant information that is based on their requirements and, more specifically, the cognitive level of their learning objectives [2].

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We performed a mapping between the course's learning objectives and the levels of Bloom's Taxonomy to concertize the epistemological position of the pedagogical aspect of the course [3]. Still, there remains a gap in our understanding of how to categorize courses pedagogically thus according to their cognitive complexity.

The field of E-learning is where the cognitive classification of text is most frequently applied. There has been a lot of attention in recent years n classifying e-learning resources using Bloom's taxonomy. Researchers use a range of techniques, from deep learning to machine learning, for automatic classification. Recurrent neural networks (RNN) perform better than conventional machine learning models, according to the findings of comparable studies. The information about text sequentiality cannot be captured by the later. While the architecture of RNNs allows them to simulate the sequential information found in the text. The benefit of machine learning algorithms is, however, how simple it is to tune them. RNNs, on the other hand, frequently overfit to tiny datasets. RNNs were outperformed by the BERT transformer. Transfer Learning and the attention mechanism are to thank for this.

In a previous article [4], by extracting features from a modified TF-IDF-POS and applying suitable weights to key terms in the learning objective based on Part-Of-Speech (POS), we provided a method for classifying learning objectives automatically. Finally, we merged those features with various classifiers. The models in combination with TF-IDF-POS produced the classification results with the best degree of accuracy. According to our research, using the suggested technique enhances the functionality of machine learning models. The accuracy of 89.17% has to be improved, though. Thereafter, in a second study [5], We proposed the BiGRU-attention, a model built using a mixture of the BiGRU algorithm and the attention mechanism. Word2vec embeddings were input into the architecture. To show the recommended model's effectiveness, we put it up against four commonly used classical machine learning algorithms for text classification that are cognitive: Bayes naive, K-nearest neighbors, Support vector machines, logistic regression, and GRU. In a third article [6], we tackled the fine-tuning of BERT by adding different layers. The outcomes showed that performance was not enhanced by adopting a more complicated layer. Additionally, we showed that the maximum accuracy value of 92.5% was achieved by the usage of dense layers on top of BERT, dropout, and the activation function ReLU. However, using BERT on a small dataset involves an overfitting problem if the model runs on multiple epochs. which is a limitation of our previous work as well as other similar works that have adopted BERT for the classification of questions [7,8] or Learning Objectives (LOs) [9]. were based on existing datasets of questions, the first dataset is composed of 600 questions [10] on which accuracy reached 87.5%, the second dataset is composed of 141 questions [11] on which the accuracy reached 70.4%. Problems with generalization and overfitting have arisen due to the small dataset sizes and variability of the training data made up of questions related to the LOs to be identified.

In this article, we take a new path for the cognitive classification of text. First, to overcome the problem of the dearth of annotated datasets of LOs, We began by gathering annotated LOs from academic sources like [60]. A total of 2400 LOs make up the training dataset. Second, we propose MTBERT-Attention, a unique model based on MTL and the mechanism of co-attention, which simultaneously learns cognitive classification with a relevant secondary task. By using auxiliary tasks, MTL effectively increases the sample size we use to train our model and helps avoid overfitting by adding inductive bias, as training is done from multiple smaller datasets. Instead of a single large dataset. Using MTL in this way improves the generalization performance of all tasks and allows data augmentation, It addresses one of our research work's shortcomings, namely the absence of annotated data [12].

The use of MTL has increased significantly in recent years. It has been adopted, together with BERT, in several studies in the field of text classification [13]. However, our article is the first to tackle the use of MTL in the field of cognitive classification. The BERT model, which is enhanced and employed as a layer to transmit knowledge between numerous tasks, is another crucial part of our MTBERT-Attention model. Despite the success of transformer-based models in several fields, their potential has not been sufficiently investigated in the task of cognitive classification. In addition to MTL and BERT, we also adopt the co-attention mechanism to optimize the efficiency of our model by allowing it to put particular emphasis on the significant aspects of the input and ignore the less important elements. To our knowledge, MTBERT-Attention is The first cognitive text categorization model, which uses an MTL Framework, a pre-trained and refined language model, and a co-attention mechanism. It surpasses in terms of performance the other models that we have adopted as Baseline.

Research works that have proposed deep learning models or BERT-based models have focused their interest only on improving classification results instead of exploring possible explainability techniques. Since these models are thought of as "black boxes," it is crucial to suggest strategies for making them comprehensible, since a teacher or learner must comprehend why the suggested course matches to the cognitive level of his research. In this sense, in addition to proposing a powerful model, we propose an explainability framework based on the attention mechanism. Our goal is to provide an explainable model, whose predictions can be explained and justified. In addition, explanations that accompany recommendations have been shown to increase user confidence and improve user satisfaction [14].

Below is a list of the contributions made by our paper:

(1) We introduce MTBERT-Attention, a unique model based on MTL, BERT, and the attention mechanism that learns the cognitive classification of texts alongside the classification of relevant subtasks. We conduct extensive experiments on our dataset with different basic models. To measure the influence of each component, we conduct an ablation experiment by adding or eliminating various aspects from the design of our suggested model.

(2) We present our explainability framework based on the attention mechanism. We conduct experiments to explore if attention scores can be used as a basis for explaining the cognitive classification task. We conduct a qualitative and

quantitative analysis of the quality of the explanations as well as the computational cost to demonstrate the effectiveness of our explainability framework. We adopt LIME [15] as the Baseline for comparison.

Our experiments show, on the one hand, that MTBERT-attention outperforms single and multi-task baselines in terms of accuracy, f1-score, precision, and recall. However, in terms of fidelity and computing cost, our approach for explainability performs better than the LIME explainer.

The remainder of this article is structured as follows: part 2 includes relevant work. part 3 describes the proposed approach. part 4 demonstrates the experiments. part 5 discusses the results and answers the different research questions of our study and part 6 is a conclusion.

Data Availability Statement

As stated in 4., along with MTBERT-Attention, we implement, train, and evaluate all the considered models. After acceptance, we plan to share all the code, training weights, datasets, and configuration.

Related work

The associated works were divided into four subsections. The first one discusses research on text classification using cognitive processes. The second one describes works using MTL and BERT for text classification. The third one describes the use of attention to enhance classification models. The last one introduces the different types of explainability methods and discusses the use of the attention mechanism as an explainability technique.

Cognitive text classification

The cognitive classification of text is mainly used in the field of E-learning. Several studies use machine learning to categorize E-Learning materials according to Bloom's Taxonomy levels. Researchers like [10,16–20] had used a combination of traditional classifiers and a combination of feature extraction techniques. The model proposed by [19] reached the highest accuracy of 86%.

Evaluation questions are the E-learning resources for cognitive classification that have received the most research. Other material kinds have, however, been addressed in certain works. For instance, [21] sought to automatically label forum posts using Bloom's taxonomy categories. [22] put out the idea of representing each specific book's content in accordance with Bloom's cognitive domain taxonomy.

The majority of earlier publications in this field evaluated the proposed approach using two open-domain datasets. The first dataset with 600 open-ended questions, which was proposed in [10]. A second dataset, which was gathered from numerous sources, books, and earlier studies [11]. There are 141 open-ended questions in it. The datasets are categorized and labeled into six groups (Knowledge, Comprehension, Application, Analysis, Synthesis, and Evaluation).

Based on Bloom's taxonomy, some research studies have classified the E-Learning material using a deep learning approach. On their part, [23] constructed a model that classifies test items using a feature-reduction-based artificial neural network (ANN) strategy. In the same sense, [24] created a model based on ANN and TF-IDF to evaluate students' cognitive abilities and capabilities. Their proposed model reached an accuracy of 85.2%. The cognitive classification of online learning content has drawn more attention in 2021. Numerous RNN-based models have been put out, [25] utilized an LSTM model to categorize the questions and learning objectives. They manually annotated the Sukkur IBA University data collection. Additionally, they utilized the 600-question dataset from previous research [10,11]. Due to the imbalance of the dataset and the difficulty in generalizing and scaling the results, this work has several limitations. In a similar vein, [26] utilized CNN and LSTM to categorize the summative evaluation. They collected 844 questions for the field of software engineering from test papers from various academic institutions to serve as their dataset. 66.67% and 44.0% are respectively the accuracies of the CNN and LSTM models. As for them, [27] proposed a bidirectional gated recurrent convolutional neural network (CNN) combined with an attention mechanism (AM-BiGRU-CNN). Their goal is to assess learners' cognitive capacity in real time while they are learning online. As a dataset, they used crawled discussions. They eliminated the discussion posts that were not cognitive before choosing the authentic interactive text data that had been submitted by 9167 students. On these data, they manually label them and run data preparation activities, which is tedious, and might result in annotation errors, which is one disadvantage of this work. Another problem is the model's substantial domain dependence on the training data set. Their suggested model ignores fine-grained language properties like grammatical norms and only relies on the automatic extraction of text features by the hybrid deep neural network. Their model's accuracy is 84.21%, although it can yet be improved.

In our previous work [6] We use a modified version of the BERT model to categorize MOOCs according to their learning objectives, other studies have adopted BERT to classify questions [7,8] and LOs [9]. In 2020, [7] used BERT and latent Dirichlet allocation (LDA) for multi-class text classification of questions. Their findings showed that BERT beat LDA, which had an accuracy of 83%, with a score of 89%. In the same sense, [8] used Google's BERT to classify assessment tasks. They used the Canterbury Question Bank as dataset. However, the question bank's distribution of questions is unbalanced, which introduces a class-imbalance problem. Moreover, due to the limitation of the dataset, the model runs only for three epochs. Because if the model runs for more epochs it tends to show signs of overfitting. On their part, [9] used BERT to classify learning outcomes. To evaluate the proposed approach, they used the same two open domain datasets used earlier [10,11]. However, the use of question datasets instead of a LOs dataset leads to a problem of heterogeneity. When a different collection of LOs is utilized to verify and test the model trained on a dataset of questions, the models are unable to generalize. The model's accuracy, which is 87.50%, can yet be improved.

Transfer Learning and MTL for text classification

The basic goal of machine learning is to improve model performance. To do so, we work on each model individually, fine-tuning it until it achieves its best results. However, the inductive bias is neglected in this technique of learning isolated tasks. On the other hand, MTL functions as a regularizer and can reduce inductive bias. Moreover, MTL can enhance the performance and generalization of classifiers by using the knowledge stored in the networks of all connected tasks [28]. This can help learn faster and more precisely and can lead to learning difficult tasks that cannot be learned in isolation. On the other hand, MTL helps with implicit Data augmentation by effectively increasing the sample size used for training the model [29].

MTL has demonstrated impressive generalization performance results on many tasks. However, combining multiple user-defined tasks does not necessarily establish better results than single-task learning and can lead to a negative learning problem. This problem is because different tasks learn at different rates and have interfering gradients so if one task is more dominant, optimization is more difficult [30]. This is because there is still no concrete understanding of the rules that share information between tasks [12]. To leverage knowledge and information across different tasks, studies have conducted experiments on various tasks and network architectures on various datasets. These studies suggested that MTL requires a careful selection of task pairs and weighting strategies to match or exceed the performance of single-task learning. These studies are based on three approaches (1) The choice of auxiliary tasks (2) The strategies of Loss weighting (3) The fine-tuning of architectures.

(1) The choice of auxiliary tasks: When a task dominates training, gradients with relatively large amplitudes are a necessary expression of this dominance. Therefore, it's important to choose the appropriate auxiliary tasks in MTL. Recently, machine learning has paid particular attention to this issue, and a number of methods for choosing the best auxiliary tasks for a given primary job have been suggested [31], and [32].

(2) Loss weighting strategies: In MTL, we try to optimize several loss functions simultaneously by combining them into a single value. In this sense, two main methods have been adopted. The first method consists in making a uniform sum of the different losses of the different tasks, in this case, the weights are, either all equal to one or chosen intuitively according to the analysis of the loss functions of the different tasks. The second method is to calculate the weights dynamically and thus solve the problem of conflicting gradients that degrade the performance of the model [33,34].

(3) Fine-tuning of architectures: In parallel with the adoption of strategies for the choice of auxiliary tasks and the choice of the loss function, MTL Studies have been based on fine-tuning of the architecture. There are two classic MTL architectures: hard parameter sharing and soft parameter sharing. Hard parameter sharing is the most adopted approach in MTL [35]. It consists in sharing the hidden layers between all the tasks; each of the tasks keeps a task-specific output layer. This approach effectively reduces the risk of overfitting. Because by training several tasks in parallel, the model captures a richer representation of the features. On the other hand, in soft parameter sharing, each task has its model with its parameters. The distance between the parameters of the model is then regularized to encourage the parameters to be similar; this regularization is done by optimizing a loss function shared between the different tasks.

For its part, transfer learning and in particular BERT has demonstrated its performance in the field of text classification. The idea of transfer learning is to train a model in one domain, and then leverage the acquired knowledge to improve the model's performance in another domain. For example, [36] demonstrates how BERT can be improved in this way to produce cutting-edge models for a variety of purposes. Recently, studies have been based on the combination of BERT and MTL to propose architectures that are more efficient for MTL. In this sense, [37] proposed the architecture (MT-DNN) which is a combination of BERT with MTL on the nine GLUE tasks. This architecture is based on "Hard parameter sharing". On their part, [13] study a multi-task learning model with multiple decoders on varieties of biomedical and clinical natural language processing tasks, Their empirical findings show that the MTL fine-tuned models perform better than cutting-edge transformer models [38]. use a shared-private MTL configuration where the shared layer is urged to learn the task-invariant features while the private layers seek to learn the task-specific representations. The gate merging method combines public and private data. The classification layer of each task receives the joint representation of each task as a final input.

Co-attention mechanism for text classification

The attention mechanism is the weighting of a huge amount of input data in such a way that the model can focus particularly on the significant aspects of the input and ignore the less important ones. Bahdanau was the one who first proposed the attention mechanism [39] for neural machine translation. The mechanism is particularly useful for machine translation since the most relevant words for the output are often found at similar positions in the input sequence. Xu [40] draws on this study to propose an attention-based model that automatically learns to describe the content of images, its model pays equal attention to all parts of the image during the caption generation process and demonstrates good results in performance. For his part, Luong [41] proposes two extensions of the attention mechanisms initially proposed by Bahdanau [39] and Xu [40]. A first extension is a global approach in which all source words are taken into account and a second extension is a local approach in which only a subset of source words is taken into account at a time. The first approach resembles the Bahdanau model but is architecturally simpler and computationally less expensive. The second approach is differentiable unlike the "Hard attention" model proposed by Xu, which makes it easier to implement. Later [42] modifies the traditional attention mechanism and proposes the self-attention mechanism which calculates the response to one position in a sequence by attending to all positions.

A co-attention mechanism is a particular form of the attention mechanism that jointly explores the attention of several tasks. [43] propose "TCAN" a co-attention network that jointly models two tasks: content and topic information. The model learns content representations based on a two-way LSTM model and builds a matrix of topic words, and combines them through the co-attention mechanism. The experimental results show that the integration of the co-attention mechanism gives an improvement of more than 13.6% in the F1 score compared to standard methods based on LSTM. [44] propose an MTL model that combines BERT with the mechanism of co-attention for the identification of dementia. They are based on two tasks: the main task for the identification of dementia and an auxiliary task for the identification of the severity of dementia. The proposed model obtains an accuracy equal to 84.99% in the detection of patients with dementia. On their part, [45] adopts the mechanism of co-attention combined with LSTM for learning mutual semantics between two tasks: Text and emojis on microblogs. The experimental results show that the proposed method outperforms several baselines for sentiment analysis. In the same direction, [46] uses the mechanism of co-attention combined with two pre-trained language models (GloVe and BERT) to capture the semantic correlations between aspect and context. Experimental results show that the proposed method can effectively achieve higher classification accuracy than other models. Co-attentive architecture with label embedding is what [47] proposes as "CNLE." They want to find representations of both the label sequence and the text sequence that can work together. to concentrate on the crucial text and label elements in order to simplify text classification. The results of the experiments show how much better their approach is than the competing methods. According to the analysis of this research work, the integration of the co-attention mechanism into MTL models has increased their performance. In our study, we propose to integrate it into our model for the cognitive classification of text.

Explainability of text classification

Explainability and interpretability allow end-users to understand, validate and trust the results created by machine learning models. Since the majority of those models are black boxes, there is a growing interest in eXplainable AI (XAI) methods. The explanation is the production of a decision in a specific instance based on interpretable features. An explainable model is preferred over an inexplicable model with the same performance [48]. Deep learning networks, made up of trillions of parameters, are the text classification models that perform the best. The reasons why some features are preferred during training over others or how correlations in training data are reflected in the choice of characteristics are mostly unknown because these networks operate like opaque black boxes. As a result, describing such intricate patterns computationally is difficult and expensive.

There are various approaches to explaining a text classification [49]. They are either model-dependent or model-independent. For less transparent models, an alternative approach is to use model-specific (or model-dependent) explanations, that is, explanation methods designed specifically for a certain type of model. On the other hand, model-independent (or model-agnostic) explanations are methods that can explain the predictions of any model. With such explanations, we are not limited to a specific type of model. These methods include LIME [15], gradient-based explanations [50], and the perturbation-based explanations of Shapely [51].

In the recent literature, various works propose using the mechanism of attention as a method of explainability [46,52]. Various works focus on attention layer weights in transformers [53] or other types of networks, such as recurrent or convolutional networks, to highlight the words or n-grams of the text most relevant to the decision. Although there are existing works on the use of the mechanism of attention as a method of explainability. This area of research is still in its infancy, and there are still pros and cons. Moreover, different researchers have different perspectives to solve the problem. So far, the academic community still lacks a unified understanding of model interpretability, there is still plenty of room for improvement.

Methodology

Bloom's taxonomy cognitive levels

A number of theoretical frameworks have been used to evaluate and categorize MOOCs [54]. believed that in order to achieve successful learning, one must adhere to the qualities of good learning [55]. This idea served as the foundation for both 7Cs as and the 12-dimensional framework as assessment framework. On the other hand, Merrill [56] claimed that all current pedagogical models and theories were built around the fundamental principles of instruction he offered. [57] created four more concepts concerning teaching material on top of Merrill's initial five principles of instruction. [54,55] and [57] based their research on the creation of assessment frameworks that were centered on open-ended questions and required an expert's help on these two theoretical frameworks. [58] employed a web-based learning methodology that guided their evaluation of MOOCs. Information, education, and learning were three global design characteristics that were included

Table 1
Our list of Bloom's Taxonomy action verbs (BTAV).

Cognitive level	Unique Action verbs	Some overlapping action verbs
Knowledge	Cite, define, enumerate, familiarize, label, list, match, memorize, name, outline, recall, recite, record, repeat, reproduce, state duplicate, quote, read, tabulate, copy, draw, underline	Explain, describe, identify...
Comprehension	Convert, discuss, explain, express, extend, generalize, paraphrase, predict, recognize, report, restate, review, rewrite, summarize, translate, associate, characterize, give examples, indicate, represent, clarify, extrapolate, give, interpolate, articulate, observe	Distinguish, describe, interpret, compare...
Application	act, apply, calculate, compute, demonstrate, dramatize, employ, illustrate, manipulate, operate, practice, schedule, show, sketch, solve, use, change, complete, backup, implement, interview, paint, utilize, adapt, simulate	Use, prepare, choose...
Analysis	analyze, categorize, contrast, diagram, differentiate, discriminate, divide, examine, point out, question, relate, separate, subdivide, test, breakdown, correlate, deduce, dissect, prioritize, survey, break, detect, diagnose, figure, inspect, inventory, investigate, debate, group	Use, explain, distinguish, compare, choose, appraise...
Evaluate	argue, assess, conclude, critique, estimate, evaluate, judge, manage, reconcile, set up, synthesize, decide, grade, weigh, counsel, mediate, predict, prescribe, probe, release, supervise, verify, attach, core, determine, value	Explain, interpret, Identify, compare, choose...
Synthesis	arrange, assemble, combine, compose, create, design, develop, devise, formulate, generate, invent, organize, plan, rate, revise, write, compile, facilitate, hypothesize, integrate, originate, propose, rearrange, role-play, improve, make, specify, tell / tell why, collect, reconstruct, reorganize	Explain, prepare...

in this more all-encompassing approach. Given the open-ended nature of the frameworks used in these investigations and the need for expert involvement, our study seeks to make learning objective evaluation automatic.

Our investigation's main purpose is to evaluate and classify MOOCs' educational components in light of their intended learning outcomes. As a result of the identification of the many cognitive learning levels and the ability to categorize learning objectives into six hierarchical levels, Bloom's taxonomy's theoretical underpinnings are most suited to our environment. It was created to help teachers create rubrics and assess their student's progress toward their learning objectives by offering rules. Developed by Benjamin Bloom, Bloom's Taxonomy is a widely used system for categorizing thought patterns [59]. Krathwohl [60] advocated that the original taxonomy be revised. A two-dimensional framework of cognitive and knowledge processes was constructed. The first dimension takes the sub-categories of the first level of the original taxonomy; the second dimension takes the six levels and renames them verbs. Due to this, the new taxonomy's nomenclature of "Remembering, Understanding, Applying, Analyzing, Evaluating, Creating" is frequently interchanged with the old taxonomy's nomenclature of "Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation" in existing research works.

Benjamin Bloom created a taxonomy of measurable verbs to help in the description and classification of learning objectives. Table 1 presents the list of Bloom's taxonomy action verbs (BTAV) that we adopted in our work. Some verbs belong to several levels which leads to confusion.

Training strategy

According to our review of the literature and our knowledge, there is no research around the combination of BERT, MTL, and the mechanism of attention for the proposal of a powerful model of cognitive classification of texts. Our strategy is to focus on choosing an optimal architecture for our model. However, before starting our experiments, we must decide on two essential points: the choice of the auxiliary tasks and the choice of the loss function.

The choice of auxiliary tasks is a research question addressed in several studies [61]. Their degree of relationship with the main task was also studied. According to [62], to increase performance, supplementary tasks in MTL setups should be linked to the primary task. In this regard, we choose an auxiliary task with the classification of emotions as its goal. On the other hand, auxiliary tasks with a uniform distribution are preferable [63], so we opt for a dataset (https://www.kaggle.com/ishantjuyal/emotions-in-text) with six classes: Anger, Fear, Joy, Love, Sadness, and Surprise. One way to gauge how similar two tasks are is by looking at how much is lost in each task. [64] show that unbalanced tasks produce highly variable gradients that can disrupt model learning. In this sense, we choose a subset of data with the same size and roughly the same distribution as our main dataset.

For the choice of the loss function, we opt for a uniform sum of the loss functions of each task as given by Equation (1).

$$L_{MTL} = \sum_{i=1}^{n} \alpha_i L_i \tag{1}$$

Where: L_{MTL} n is the number of tasks, and α is the weight of the loss function L_i for each task. As the loss function, we utilize "Categorical Cross-Entropy loss" and apply it to each task as given in Equation (2):

$$L_i = -\sum_{c=1}^{k} 1(S,c).log\left(Pr\left(\frac{c}{s}\right)\right) \tag{2}$$

in the case when c, the class label, is the appropriate classification for S, 1(S, c) is a boolean (0 or 1), and Pr(.) is defined by Equation (3), and k is the number of scalar values in the model output. This loss is a very good measure of how distinguishable two discrete probability distributions are from each other. The minus sign ensures that the loss gets smaller when the distributions get closer to each other. Given that one example can be taken to correspond to one level of cognition with a probability of 1, and to other levels with a chance of 0, the categorical cross-entropy is a good fit for our classification job. Due to the fact that our classification problem involves numerous classes, The output layer is constructed on top of a softmax activation layer. The categorical cross-entropy loss function is only compatible with Softmax as an activation function. The formula for the softmax function is:

$$Pr\left(\frac{c}{S}\right) = softmax(\vec{s}) = \frac{e^{si}}{\sum_{j=1}^{k} e^{sj}} \tag{3}$$

Where $\vec{s} = (s_1; \ldots\ldots; s_K)$, s_i are the components of the multi-class classifier's input vector, and k is the number of classes. The predicted label for the input is then determined by selecting the output node with the highest probability. To determine the order of instances in an epoch. We adopt "Forced heterogeneity" As an approach for Epoch Sampling, and the "Uniform Batches" strategy, which is based on an equal number of instances of each task in every batch [65]. Algorithm 1 describes this strategy.

Algorithm 1: Training the MTBERT-Attention model.

Data: Prepare the data for T tasks
Initialization:
Initialize model parameters randomly
Pre-train the shared layers
Set the max number of epochs: epochmax
For each epoch i do
    For each batch B of data do (uniform batches from the T Datasets)
        Compute loss on each task Eq (2)
        Compute loss for all tasks $L_{MTL}L(\theta)$ Eq (1)
        Compute gradient:$\nabla(\theta)$
        $Updatemodel{:}\theta = \theta - \eta\nabla(\theta)$

The proposed MTBERT-Attention model

The architecture of our model is based on the principle of "Mixed parameter sharing" and the co-attention mechanism. We combine the two architectures Soft and Hard Parameter Sharing and we add a co-attention layer to catch the attention weights and construct context vectors. Then, we concatenate the two context vectors. The overall design of our suggested model is shown in Fig. 1. The "BERT Layer," "Co-attention Layer," and "Output Layer" are the three basic components, and a global loss function is also utilized.

The BERT Layer: This is the shared layer. A Bidirectional Encoder Representation from the Transformers (BERT) model is used to encode the input learning objectives [36]. A text is first cleaned before being tokenized with BERT's default tokenizer and then converted into pre-trained BERT embeddings $''input_i$," where "i"is the task number. The pre-trained BERT model receives these embeddings. Because we must determine the attention scores for all tokens, not just the [CLS], we use the sequence output in this case rather than the pooled output.

$$Bert_{Oi} = BERT(Input_i) \tag{4}$$

The Co-attention Layer: We propose a novel mechanism that jointly reasons cognitive attention and emotional attention, which we refer to as co-attention. The co-attention layer performs a series of computations as described below. Our co-attention Layer is based on a straightforward attention mechanism that draws inspiration from Bahdanau's additive attention theory [39]. Our method takes the output of the BERT Layer as a single input parameter. This representation is first fed into the Tanh activation function to obtain U_m and U_A as hidden representations of the main and auxiliary tasks respectively. This activation function's weights and biases are initialized at random:

$$U_m = tanh\left(W_m * Bert_O^m + b_m\right) \tag{5}$$

$$U_A = tanh\left(W_A * Bert_O^A + b_A\right) \tag{6}$$

The co-Attention layer will then multiply the representations, to determine each word's measures of firm performance for a learning target with T words. We use the normalized significance weight α_t obtained from the softmax function to

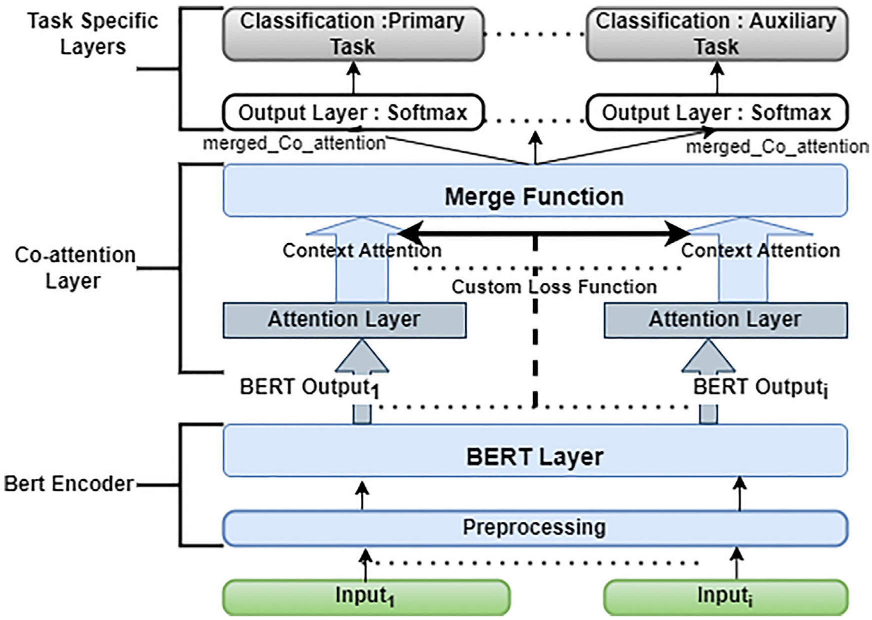

Fig. 1. The proposed MTBERT-Attention model.

create a probability distribution over the words.

$$\alpha_t^m = \frac{exp(U_m * w_a^m)}{\sum_{i=1}^{T} exp(Bert_{Oi}^m * w_a^m)} \tag{7}$$

$$\alpha_t^A = \frac{exp(U_A * w_a^A)}{\sum_{i=1}^{T} exp(Bert_{Oi}^A * w_a^A)} \tag{8}$$

Finally, using the attention significance scores as weights, we construct V, the output representation vector for the text, as a weighted summation over all the words.

$$V^m = \sum_{t=1}^{T} \alpha_t^m Bert_O^m \tag{9}$$

$$V^A = \sum_{t=1}^{T} \alpha_t^A Bert_O^A \tag{10}$$

To obtain the co-attention of the two attentions, we use a function to merge the attention context vectors and to concatenate the respective learned features for learning objectives and emotions respectively through co-attention. This representation vector obtained from the attention layer is a high-level encoding and it is used as input to the classifier.

$$merged_{Co_{attention}} = Concatenate(V^m, V^A) \tag{11}$$

The Output Layer: The relevant task-specific softmax layer receives the outputs of the concatenation phase and performs classification on them. The network's parameters are adjusted to reduce the distributions' cross-entropy across all tasks.

$$Output = softmax(W_f * merged_{Co_{attention}} + b_f) \tag{12}$$

Where $Output = [Output_{main}, Output_{Aux}]$ is the final forecast, which displays the degree of certainty for each class. Learned weights and biases are Wf and bf.

The loss function: We use the custom loss function defined in Equation (1). The loss weight values are intuitively optimized during the experimentation phase. Custom loss functions allow us to condition the loss during training on arbitrary information, and not just on the mismatch between predicted and ground truth labels. This opens up possibilities for regularizing distances between weights across layers in separate networks. For each layer, model parameters are upgraded with the backpropagation algorithm, which iteratively computes gradients using the chain rule. We employ the Adam optimizer [66] together with the low-learning-rate updates of the warm-up steps to support the convergence of the model. The number of epochs, the size of the batch, and the average length of the training data for the two jobs are added to determine the value of the warmup steps.

The proposed explainability framework based on attention mechanism

Attention weights reflect important inputs for an output task. It is interesting to utilize attention weights to explain the behavior of BERT because it is an attention-based model. BERT, however, employs different attentional mechanisms. There are 144 attention mechanisms in the basic BERT type, which consists of 12 layers with 12 attention heads each. Furthermore, because of the interconnectedness of the BERT layers, the hidden embeddings, which can be combined representations of several embeddings instead of the words are the main focus. According to recent studies, each of these heads of attention concentrates on various patterns (verbs, determiners, prepositions…). The complex modeling skills of BERT are made possible by the opaque combination of each of these several attention patterns. The decision of which mechanism(s) to use to explain the model is now a hurdle.

To benefit from the advantages of the attention mechanism, we propose in our approach (See Algorithm 2) to add another

Algorithm 2: Our Attention-based Framework Explainability.

```
Initialization:
Pre-train the BERT Encoder
Train the Attention layer
For each term in the Dataset do
    For each word in the term do
        Compute Attention tokens Eq (7) and Eq (8)
        Compute Attention scores Eq (13)
    End
End
    Compute prediction based on attention scores Eq (14)
```

layer of attention based on the same attention mechanism adopted in our MTBERT-Attention model. Additionally, as we noticed in our experiments, adding a co-attention layer increases the performance of the BERT-only model. We decide to choose to use this mechanism and not the self-attention mechanism of BERT.

Our model also serves as an explainability framework; Thanks to the attention layer, we can extract the attention scores for each word from the attention tokens. Since the BERT tokenizer splits some words into multiple tokens. For a word W for which we want to calculate the attention score and t a token that belongs to it, we remove the attention scores of the tokens [CLS] and [SEP]. The attention of a word W is calculated as in Equation (13).

$$\alpha_W = \sum_{t \in W} \alpha_t^m \tag{13}$$

To compare these explanation scores to the original black-box model's ground truth labels and predictions on the test set. We needed to make a prediction based on the explanation scores, so we created a simple classifier using a softmax activation function that computes the label of a sentence P_{scores} using the attention scores generated by the attention layer as input.

$$P_{scores} = Softmax(\alpha_W) \tag{14}$$

Experimental research

This chapter provides a thorough description of the baseline models and implementation details in addition to a thorough study of the data sets. We will in this chapter:

- Test our suggested model in experiments and contrast it with the standard models.
- Test our suggested explainability framework and contrast it with a standard XAI approach.

Datasets analysis

Main Task: The dearth of annotated datasets poses a significant problem for this research. Only question datasets [17] are available. We attempted to train using these datasets at first. However, the models failed to generalize when we used a different set of LOs to validate and test them. The heterogeneity of the data is to blame for this. Building our dataset was the only way to solve the problem. We began by gathering annotated LOs from academic sources like [67]. A total of 2400 LOs make up the training dataset. In the form of a histogram, Fig. 2 displays the distribution of LOs per class and word count for each input dataset. The top five frequent words par class are presented in Table 2.

Auxiliary Task: The Auxiliary dataset consists of 2400 training targets. We choose to limit the number of emotional texts to the same number of learning objectives. Hence, we can use the “Uniform Batches” strategy as explained in chapter 3.2. The number of classes is six. The average length of the emotions is about 120 words. The classes are balanced, with 400 emotions per class.

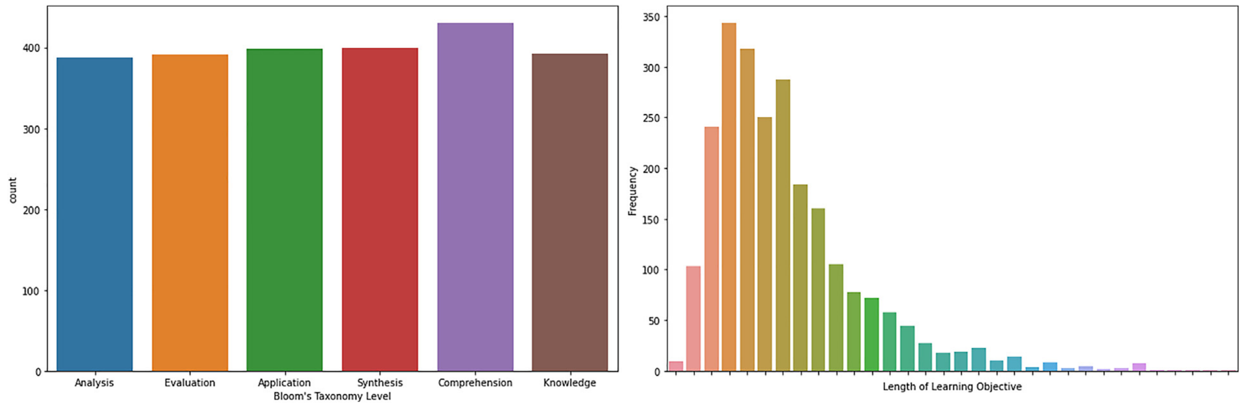

Fig. 2. The distribution of data by frequency and lengths.

Table 2
The five most common words for each level.

Cognitive Level	The five most common terms	Frequency
	analyze	117
	data	47
Analysis	differentiate	44
	compare	40
	given	39
	evaluate	80
	given	79
Evaluation	cost	57
	estimate	52
	determine	46
	apply	108
	illustrate	64
Application	demonstrate	46
	data	38
	employ	38
	explain	127
	discuss	107
Comprehension	describe	64
	data	38
	programming	33
	design	150
	given	66
Synthesis	using	64
	develop	61
	data	51
	identify	126
	define	96
Knowledge	able	64
	students	55
	describe	52

Baseline models and baseline XAI method

Performance baselines

Fine-tuned BERT [6]: In our previous study, we train several BERT-based models for the classification of our main task. These models are combined with either dense layers or classifiers like LSTM and Bi-LSTM. From our experiments, BERT combined with dense layers gives better results for the cognitive classification of text. In this architecture [6], We add layers that are all connected. The completely linked layer converts the output of the twelve BERT layers into the final output of six classes, which stand in for the six Bloom's taxonomy degrees of cognition. Three thick layers make up this stratum.

We compare the recent models proposed by [7] and [9] with our fine-tuned BERT Model proposed in our previous study. we use the datasets of questions [10] and [11].

We examine the MTBERT-Attention model with baseline models, such as those from recent studies and variations of our suggested model by adding or eliminating certain characteristics from its architecture:

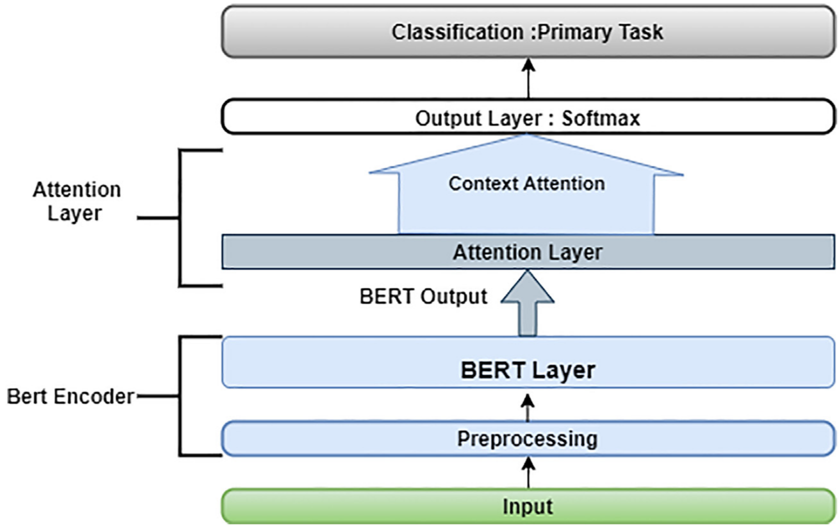

Fig. 3. The architecture of BERT-Attention model.

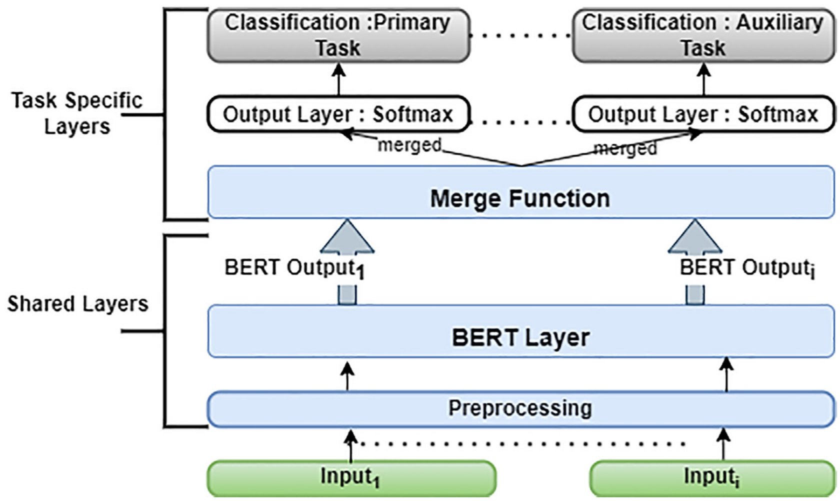

Fig. 4. The architecture of Hard Parameter Sharing MTL model.

CNN [26]:
LSTM [26]:
AM-BiGRU-CNN [27]:
BERT-Attention: We train a model (see Fig. 3) with an attention layer based on our proposed approach for the single cognitive classification task.

MT-Bert Hard: Our implementation of "Hard parameter sharing" is shown in Fig. 4 We use BERT as a shared layer structure. Then we concatenate the outputs "BERT Output$_i$" to pass this joint representation to the last layer for classification.

MT-Bert Soft: Unlike "Hard parameter sharing", we do not combine any layers from both tasks. Moreover, we transfer the regularization between the respective layers to the loss function. As shown in Fig. 5 we implement "Soft Parameter Sharing" using the previously defined loss function in Equation (2). For the weight values, we experiment with different combinations.

MT-Bert Mixed: We combine the hard and soft parameter sharing by using a shared loss function and combining the BERT outputs as shown in Fig. 6.

Explainability baseline

To measure the performance of our framework for explainability, we chose LIME [15] as a baseline for comparison. LIME is model-agnostic, it can be used with any machine-learning model, therefore. Additionally, It has the capacity to be locally interpretable, that is, to provide justifications for a single instance. LIME modifies the input data to produce a succession

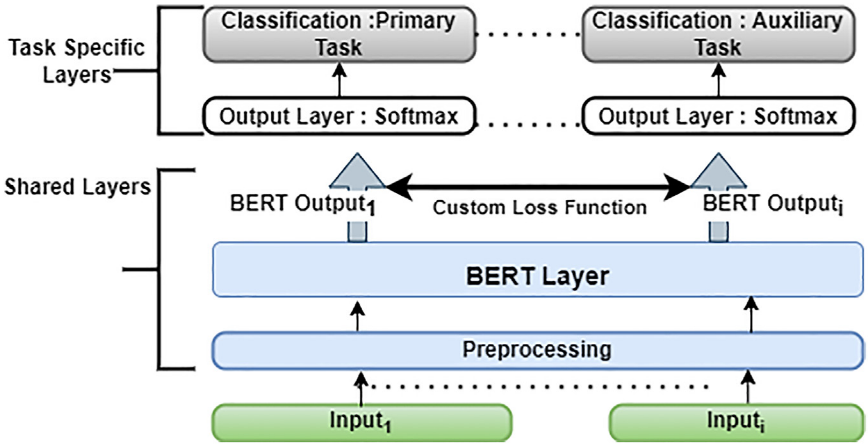

Fig. 5. The architecture of Soft Parameter Sharing MTL model.

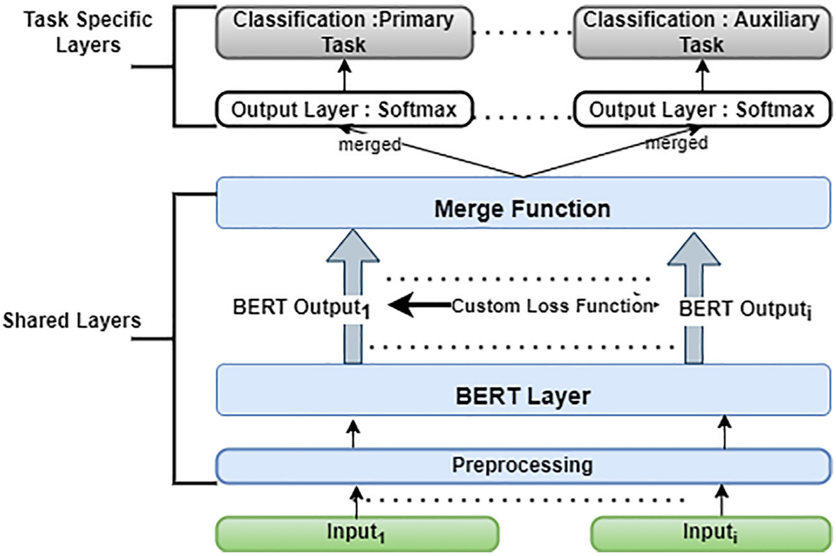

Fig. 6. The architecture of Mixed Parameter Sharing MTL model.

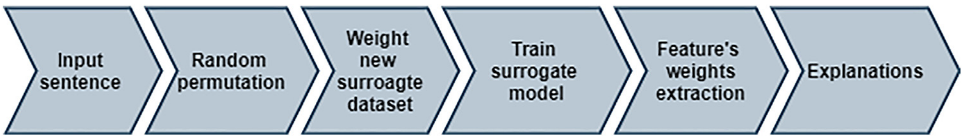

Fig. 7. LIME framework for explainability.

of false data that only partially retains the original features. As a result, in our example, LIME conducts permutations by arbitrarily removing the letter "s" from the original sentence when given an input sentence "s". Classify this new surrogate dataset using the MTBERT-Attention model, and then compute the distance between "s" and all conceivable permutations using the cosine similarity metric. Then, The similarity score is calculated by LIME using this distance pi_s. Additionally, it chooses from the permuted data m characteristics that most accurately represent the MTBERT-Attention model result. A less complex self-explanatory surrogate model is also used by LIME on the samples using only the m obtained features from the permuted data weighted by its similarity to the original observation. Finally, LIME Extracts the feature weights from the simple model and uses these as explanations for the complex model's local behavior. The entire LIME framework is presented in Fig. 7.

The learned model should be a good approximation of our original model predictions locally, but it does not have to be a good global approximation. This kind of accuracy is also called local fidelity. Mathematically, the learned model can be expressed as follows:

$$Explanation(s) = \arg\max_{g \in G} L(f, g, \pi_s) + \Omega(g) \tag{15}$$

The explanation model for input sentence s is the model g that minimizes loss L, which measures how close the explanation is to the prediction of our original model f while the model complexity $\Omega(g)$ is kept low. LIME calculates $\Omega(g)$ based on the number of features and the size of samples. G is the group of conceivable model explanations. The similarity score between the inputs from the surrogate dataset and the proximity measure π_s is defined.

As LIME is not designed for the MTL architectures, we adapted it to our context, by constructing our encoder that takes as input both LO and emotion and returns embeddings as tensors with the same dimension. Then, we fed those embeddings to our original model and returned as output an array of probabilities of the cognitive task.

Implementation and training details

To create the BERT models, we employed “Google Colab” and “Tensorflow” in addition to “Keras Tensorflow”. We utilize Tensorflow Hub (TFHub). We employ the fundamental pre-trained model $''bert\_multi\_cased\_L-12\_H-768\_A-12/2''$ for our investigations. We construct our BERT layer using the Tensorflow Hub’s KerasLayer function in order to implement the models chosen. Then, based on the variables of this layer, we tokenize our text, giving us the initial input for our BERT model. Then we created the embeddings of position $''input\_mask''$ and segments $''segment\_ids''$ which are the other two inputs of BERT.

We implement our attention layer as a KerasLayer, which takes as input the sequence output of the Bert layer and give it as outputs: The attention scores and the attention context vectors. After experimenting with numerous alternative setups and numerous fruitless attempts that ran out of memory, we discovered a working setup of hyper-parameters. 3e-5 is the base learning rate. With a batch size of 16, We empirically set our primary model’s maximum epoch count at 5, and we store the best model for testing on the validation set. We selected an appropriate number of epochs for the remaining baseline models in order to prevent overfitting and reduce loss. We divided the dataset into training and test portions for each model, 80%, and 20% respectively. To determine the appropriate number of epochs for each model, we use the “early stopping” technique with the value of the loss as a monitor, with patience equal to 3.

Evaluation metrics for Performance and explainability

The selection of the most appropriate metrics to assess how well a certain classifier performed on a particular data set is influenced by a number of factors, including class balance and anticipated outcomes. Since there are about equal numbers of samples from each class in both datasets, we can only evaluate the performance of our models using the accuracy metric [68]. Nevertheless, we also employ precision, recall, and F1 score measurements to provide a more comprehensive picture of performance. Metrics are defined as follows in a classification problem with more than two classes:

$$Accuracy = \frac{\sum_{c \in C}(TP_c + TN_c)}{\sum_{c \in C}(TP_c + TN_c + FP_c)} \tag{16}$$

$$Precision = \frac{\sum_{c \in C} TP_c}{\sum_{c \in C}(TP_c + FP_c)} \tag{17}$$

$$Recall = \frac{\sum_{c \in C} TP_c}{\sum_{c \in C}(TP_c + FN_c)} \tag{18}$$

Where: TP_c = True Positives in class c, TN_c= True Negatives in class c, FN_c= False Negatives in class c, FP_c= False Postives in class c. The F1 score can be thought of as a weighted average of recall and precision, with a best value of 1 and a worst value of 0. We use the weighted average F1 score, which is determined by averaging all F1- scores per class and accounting for each class’s support. Support refers to the number of actual occurrences of the class in the dataset. “Weight” basically refers to the proportion of support of each class compared to the sum of all support values. With a weighted average, the output average represents the contribution of each class weighted by the number of examples of that given class.

Matthews Correlation Coefficient (MCC): To measure the correlation between our model predictions and ground truth labels, we use the Matthews Correlation Coefficient (MCC), which aims to measure the strength of the relationship between two variables. The metric (MCC) was developed in 1975 by Brian W. Mattheus [68]. The confusion matrix C for k classes can be used to define the Matthews correlation coefficient in the case of many classes.

$$MCC = \frac{c * s - \sum_{k \in K} p_k * t_k}{\sqrt{(s^2 - \sum_{k \in K} p_k^2) * (s^2 - \sum_{k \in K} t_k^2)}} \tag{19}$$

To simplify the definition, we consider the following intermediate variables:
$c = \sum_{k \in K} C_{kk}$ the number of correctly predicted items.
$s = \sum_{i \in K} \sum_{j \in K} C_{ij}$ the number of items.
$p_k = \sum_{i \in K} C_{ki}$ the frequency with which class k was predicted (column total)).
$t_k = \sum_{i \in K} C_{ik}$ the real number of instances of class k (row total).

To compare our proposed explainability framework and the LIME explainer, we use the fidelity metric: This metric [49] assesses the difference between each XAI technique and the original model. Given a Dataset = X, Y, the evaluation consists of observing each pair of data records and the target value (x,y) of the dataset, the correspondences between y, and the model prediction XAI technique. Fidelity can be interpreted as the accuracy of the XAI technical model concerning the predictions of the original model.

Experimental results analysis

In this section, we examine the performance of our suggested model and the efficiency of our suggested explainability framework. Our primary goal is to investigate how well MTL, BERT, and the co-attention mechanism work together to perform cognitive text classification and to assess the effects of various tuned architectures. We employed the baseline models already presented in section (4.2.1): (1) Single-Task BERT (2) BERT-Attention (3) MT-Bert Hard (4) MT-BERT Soft and (5) MT-BERT Mixed. In the second phase, we investigate the effectiveness of our proposed explainability framework compared to our defined LIME explainer. More precisely, we hope to provide answers to the following major research topics through investigations:

Research question 1: Does the use of auxiliary tasks effectively improve the generalization performance of the cognitive task? How can the different fine-tuning architectures of our MTL framework have a different impact on the cognitive classification task?

Research question 2: How much does the co-attention mechanism strengthen the attribution of features in the outcome prediction and enhance the functionality of our model?

Research question 3: Is it possible to annotate relevant words using the attention mechanism? that drives classification? How efficient and realistic is our explainability framework based on this mechanism?

Experimental results

Tables 3 and 4 show the results of our experiments on the main and auxiliary tasks for the training and validation datasets. The greatest result, 97.71%, is displayed in Table 3 and reflects the precision of our suggested model, “MTBERT-Attention.” Single-tasking and multi-tasking baselines are outperformed by our suggested model. We observe that the “Single-Task BERT” baseline, based on BERT, can obtain good performance on the main task. It reaches an accuracy of 95.58%, which is equal to the accuracy of the “MT-BERT Soft” model for the cognitive task. But the “MT-BERT soft” model significantly improves accuracy on the auxiliary task, which goes from 88.13% for the “Single-Task BERT” model to 90%. We also notice an increase in the accuracy of the “BERT-Attention” model compared to the baseline “Single-Task BERT” and also a decrease in loss.

Among the multitasking models, we notice that the “MT-Bert Hard” model surpasses the “MT-BERT Soft” model in terms of accuracy, it also manages to minimize the loss function. “MT-BERT Mixed” reaches a loss value of 0.10 on the main task and 0.14 as the loss of the whole model which is very low compared to the loss of the model “MT-Bert Hard” which is 0.32. “MTBERT-Attention” outperforms all multitasking baselines, it manages to reach the maximum value of accuracy and also to minimize the loss for both tasks and also for the general model.

Table 3
Models Results for the training Datasets. The best results are in bold.

Model	Principal Task				Auxiliary Task				
	Accuracy	F1 score	Loss	MCC	Accuracy	F1 score	Loss	MCC	MT-Loss
Single-Task BERT	99.22%	99.22%	0.0168	0.9906	98.70%	98.70%	0.04	0.9844	-
BERT-Attention	99.48%	99.48%	0.02	0.9988	98.33%	98.33%	0.06	0.9800	-
MT-BERT Soft	99.43%	99.43%	0.03	0.9938	98.18%	98.17%	0.06	0.9351	0.0468
MT-Bert Hard	99.95%	99.95%	0.042	0.9925	99.90%	99.90%	0.04	0.9938	0.0086
MT-BERT Mixed	99.74%	99.74%	0.0173	0.9963	99.79%	99.79%	0.02	0.9956	0.0190
MTBERT-Attention	**99.95%**	**99.95%**	**0.0023**	**0.9969**	**99.95%**	**99.95%**	**0.0114**	**0.9975**	**0.0144**

Table 4
Models Results for the validation Datasets. The best results are in bold.

Model	Principal Task				Auxiliary Task				
	Accuracy	F1 score	Loss	MCC	Accuracy	F1 score	Loss	MCC	MT-Loss
Single-Task BERT	94.58%	94.59%	0.20	0.9351	88.13%	88.01%	0.61	0.8576	-
BERT-Attention	95.63%	95.63%	0.18	0.9477	89.17%	89.12%	0.68	0.8701	-
MT-BERT Soft	94.58%	94.58%	0.26	0.9351	90.00%	89.89%	0.46	0.8804	0.36
MT-Bert Hard	95.63%	95.61%	0.16	0.9476	94.00%	94.98%	0.16	0.9399	0.32
MT-BERT Mixed	96.88%	96.88%	0.10	0.9269	94.38%	94.36%	0.17	0.9275	0.14
MTBERT-Attention	**97.71%**	**96.72%**	**0.08**	**0.9994**	**94.38%**	**94.37%**	**0.19**	**0.9994**	**0.14**

Table 5
Results of our cognitive task classification of the three models for the test datasets. The best results are in bold.

Model	MTBERT-Attention			MT-BERT Mixed			BERT-Attention			
	F1 score	Precision	Recall	F1 score	Precision	Recall	F1 score	Precision	Recall	Support
Analysis	95.71%	91.78%	100%	94.96%	91.67%	98.51%	95.71%	91.78%	95.71%	67
Application	**100%**	**100%**	**100%**	**98.72%**	97.47%	**100%**	**97.40%**	97.40%	**97.40%**	78
Comprehension	97.75%	97.75%	97.75%	96.05%	96.59%	95.51%	94.92%	95.45%	94.92%	86
Evaluation	97.56%	98.77%	96.39%	97.56%	**98.77%**	96.39%	94.61%	94.05%	94.61%	85
Knowledge	98.32%	98.88%	97.78%	97.21%	97.75%	96.67%	94.70%	95.65%	96.70%	93
Synthesis	96.55%	98.59%	94.59%	96.55%	98.59%	94.59%	94.29%	**100%**	94.29%	71

Table 6
Some misclassified learning objectives. BTAV verbs are highlighted in gray.

Learning Objective	True Label	Predicted Value		
		MTBERT-Attention	MT-BERT Mixed	BERT-Attention
Prepare a case to present and evaluate your view about	Synthesis	Synthesis	Synthesis	Evaluation
Given an application what to choose between abstract classes and interfaces	Evaluation	Evaluation	Application	Evaluation
Examine need for pilot system and classify the types of pilot	Analysis	Analysis	Comprehension	Comprehension

We also notice that the values of the F1-score are very close if not equal to the values of the accuracy of the model, this was expected since our data set is quite balanced.

The MCC values are not positively correlated to the accuracy values, we notice that the single-task models have higher MCC values than the "MT-BERT Mixed" multi-task model which has higher accuracy. The MCC value of the "MT-BERT Hard" model is higher than that of "MT-BERT Mixed" even if the latter is more efficient in terms of accuracy.

Results for the three models "MTBERT-Attention" were analyzed for each class, "MT-BERT Mixed" and "BERT-Attention" (see Table 5), we note that our model is more efficient in the classification of the cognitive level "Comprehension" compared to the other two models. Although the "MT-BERT Mixed" model classifies learning objectives at the "Application" cognitive level with the best f1-score and recall, it is more precise for the classification of cognitive level "Assessment" learning objectives. We also notice that our model improves the precision and the recall of all cognitive levels except the precision of the "Analysis" level which deteriorates, on the other hand, the recall improves and positively impacts the f1-score.

In terms of all the metrics adopted, our suggested model achieves ideal values. Nevertheless, 0.08 is not an ideal value for cognitive task loss. This is because "outliers" create errors that are expensive because the predicted probability is quite severe, even if the model is effective and makes few mistakes. In the case of categorical cross-entropy, the precision measures the true positive, i.e. the precision of discrete values, while the log loss of the softmax loss is as it were a continuous variable that measures the model performance against false negatives. A wrong prediction slightly affects the accuracy but penalizes the loss disproportionately. We continue to analyze the results in order to better understand the cause of these problems. According to the confusion matrix of our model (see Fig. 8), the "Analysis" cognitive level is the least precise with a rate of 13.75% of correct predictions. The majority of "Synthesis" cognitive level learning objectives that are misclassified are in the "Analysis" class. To understand what drives our model to confuse certain cognitive levels, we push the analysis of misclassified learning objectives.

From Table 6 as well as the visual and manual analysis of the learning objectives, we find that the majority of prediction errors of the three baseline models "MTBERT-Attention", "MT-BERT Mixed" and " BERT-Attention" come from the confusion caused by certain verbs in Bloom's taxonomy. From the results, it was observed that a set of action verbs from Bloom's taxonomy were truly ambiguous because it is difficult to identify the required cognitive level unless the context is known. Examples of these action verbs from Bloom's taxonomy are presented in Table 1.

Table 7 represents some learning objectives misclassified by our proposed model. We can see from a visual analysis of the words of each learning objective that these errors are also caused by the confusion involved in using action verbs from Bloom's taxonomy. That said, the analysis based on the verbs of the BTAV list is not sufficient since our model also takes into account the context and the attention attributed to each token of the learning objective. Further analysis of the explainability and interpretability of our model requires the adoption of an adequate framework.

Ablation study

To measure the impact of each component, we undertake an ablation study. In this study, several features from the architecture of our suggested model are added or removed. The already-presented fundamental models are the updated versions.

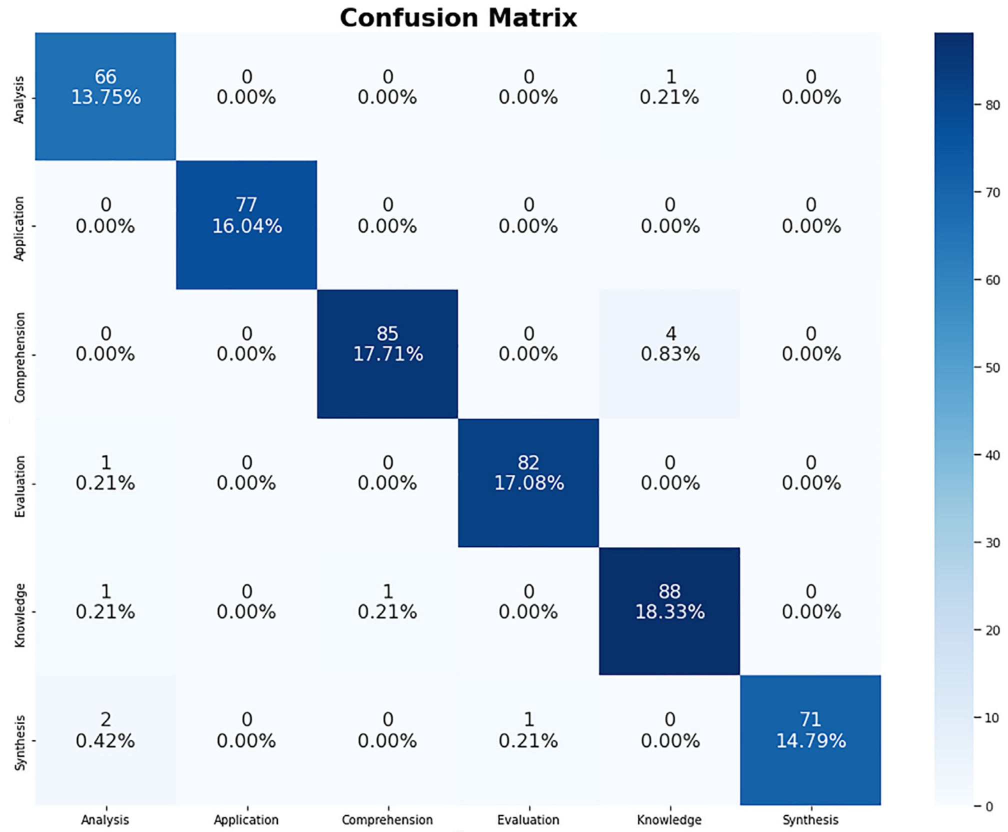

Fig. 8. MTBERT-Attention Confusion Matrix.

Table 7
MTBERT-Attention predictions for some learning objectives with overlapping action verbs.

Learning Objective	True Label	Predicted Value
On successful completion of this course the students will be able to familiarize with various land economic concepts	Knowledge	Analysis
State timing diagram machine cycle and t state and analyze the need for timing diagram	Analysis	Knowledge
Apply data cube computation method to determine suitable path	Application	Evaluation
Identify the requirements of diverse applications with help of object oriented software engineering approach	Comprehension	Knowledge
For a given software application develop a test plan to determine whether a product works with the intended users	Synthesis	Evaluation

Effectiveness of MTL

In this subsection, we answer research question number 1. We evaluate the MTL models against the “Single-Task BERT” and “BERT-Attention” models.

The choice of the auxiliary task: We observe a positive and mutual impact between the two auxiliary tasks. The classification performance at the two-task level increased when adopting MTL models compared to single-task models where each task is trained separately. We conclude that our choice of the auxiliary task is judicious and that the correlation between the cognitive task and the emotional task is strong.

The choice of custom loss function: We are trying to optimize two loss functions, which requires a way to combine these loss functions into a single value. To optimize the loss function defined in Equation (1), we intuitively try several values for the loss weights, the optimal values of the weights for our proposed model “MT-BERT Attention” according to our experiments are (0,8.0,2). Our loss function is therefore defined as follows:

$$L_{MTL} = 0.8 * L_1 + 0.2 * L_2 \tag{20}$$

Where L_1 is the loss function of our main task and L_2 is the loss function of our auxiliary task. We use Equation (2) for the calculation of both loss functions. We adopt this custom loss function in MTL models except for the "MT-Bert Hard" model which is based on hard parameter sharing only.

For the "MT-Bert Soft" model, the loss function is defined as follows:

$$L_{MTL} = 0.5 * L_1 + 0.5 * L_2 \tag{21}$$

By comparing the "MT-Bert Soft" model to the "Single-Task BERT" model, we find that accuracy has not improved. We also find that the MTL was more beneficial for the auxiliary task than the cognitive task, this is perhaps due to the choice of the weights of the global loss function. But the loss value of each task has improved. That said, the adoption of soft parameter sharing based only on the optimization of the loss function does not improve the general performance of the model. This pushes us towards the adoption of hard parameter sharing as an MTL architecture.

Fine-tuning of architectures: the main objective of our research is to propose an optimal architecture. For this, we study several possible architectures. We find that soft parameter sharing is not enough. So we test hard parameter sharing by building an MTL model architecture where both main and auxiliary tasks share the same layers but each retains an output layer. We notice that this architecture improves the accuracy, the f1-score, and the MCC and also the loss of the cognitive task which goes from 0.26 to 0.16, and the emotional task which goes from 0.46 to 0.16. The overall loss of the "MT-BERT hard" model which is 0.32 remains higher than the loss of the "MT-BERT soft" model which is 0.36. Such a high loss means that our model will not generalize well, so we propose another architecture. We build the "MT-BERT Mixed" model which is a combination of both parameter-sharing techniques. This model improves performance with an accuracy of 96.88% and significantly reduces the losses of each task and also the overall loss which drops from 0.32 to 0.14.

From the results of our experiments, We note that the performance of the classification model is significantly impacted by the co-training of our primary task with a secondary task. This demonstrates that MTL is especially helpful for situations where there is a dearth of training data.

Effectiveness of co-attention mechanism

In this subsection, we answer research question number 2. To explore the effect of the co-attention mechanism in our proposed approach, We contrast the "MT-BERT Mixed" and "BERT-Attention" models' architectures with that of our "MTBERT-Attention" model. We find that the MCC of the two models with attention is higher than that of the "MT-BERT Mixed" model, which means that the models with attention are stronger in the correlation between the predicted values and the true label values.

To further explore the effect of the co-attention mechanism in our proposed method, we compare the architecture of the "MT-BERT attention" model to several modified attention models. In "MT-BERT Mixed", we remove the attention module and take the output of the last cell of the BERT layer as the representation. In "BERT-Attention", we use the attention mechanism based on the main task of cognitive classification without the adoption of MTL. In "MT-BERT attention", we extend the "MT-BERT Mixed" model by adding a co-attention layer and concatenating the output of the co-attention module to a fully connected layer with a softmax function as a classifier. Table 4 shows the improvements in accuracy compared to baselines, which confirms that the co-attention module plays an important role in our model.

Co-attention (fusion of the two vectors of attention) models the interactions between emotions and learning objectives and helps to reduce the model's attention span to significant parts of the input, thereby reducing space model search. Emotional attention to the learning objective is useful to filter out the error accumulation problem caused by error prediction. The features are combined in the form of concatenation, which highlights the auxiliary effect of the model structure on the feature extraction and helps to capture the contrasting features between the channels so that the model can strengthen the categorization by better understanding the word's context through the use of neighboring terms.

Table 8
The number of matches between our proposed explainability framework and the LIME explainer.

Model	A	L
G	2378 (99.08%)	2339 (97.46%)
T	2372 (98,83%)	2329 (97.04%)

Table 9
Three case studies for the comparison of the scores of the two explainability techniques.

Case	Learning Objective	True Label	Predicted Value	
			BERT-Attention	LIME
First case	Examine heuristics to generate compact set of itemsets	Analysis	Analysis	Analysis
Second case	Ability to outline the steps involved in handling interrupts by i/o software and the functions of device independent i/o software	Analysis	Analysis	Knowledge
Third case	Compare the features of the data warehouse with a data mart with respect to querying granule size and storage space parameters	Analysis	Evaluation	Evaluation

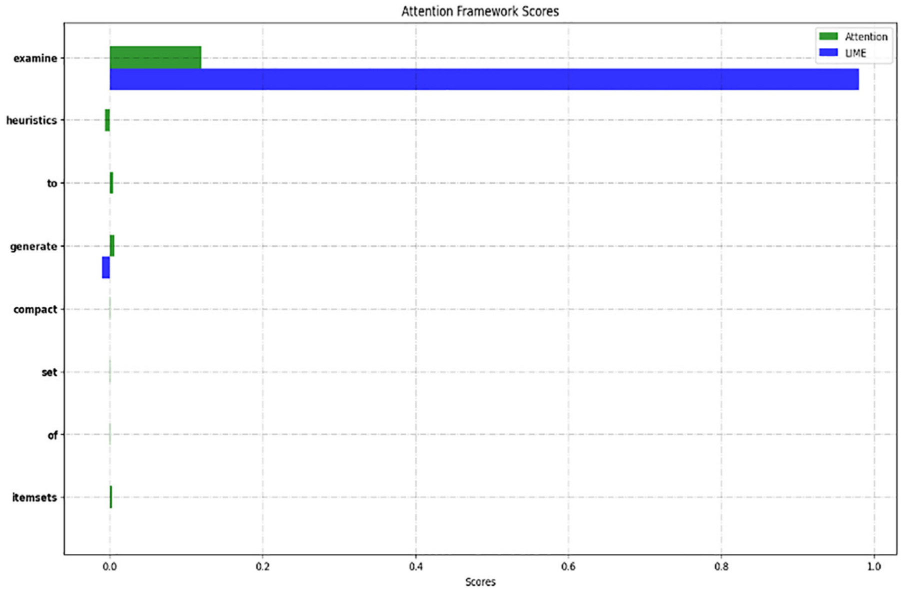

Fig. 9. Comparison of score assignment of LIME explainer and attention explainer for the first case.

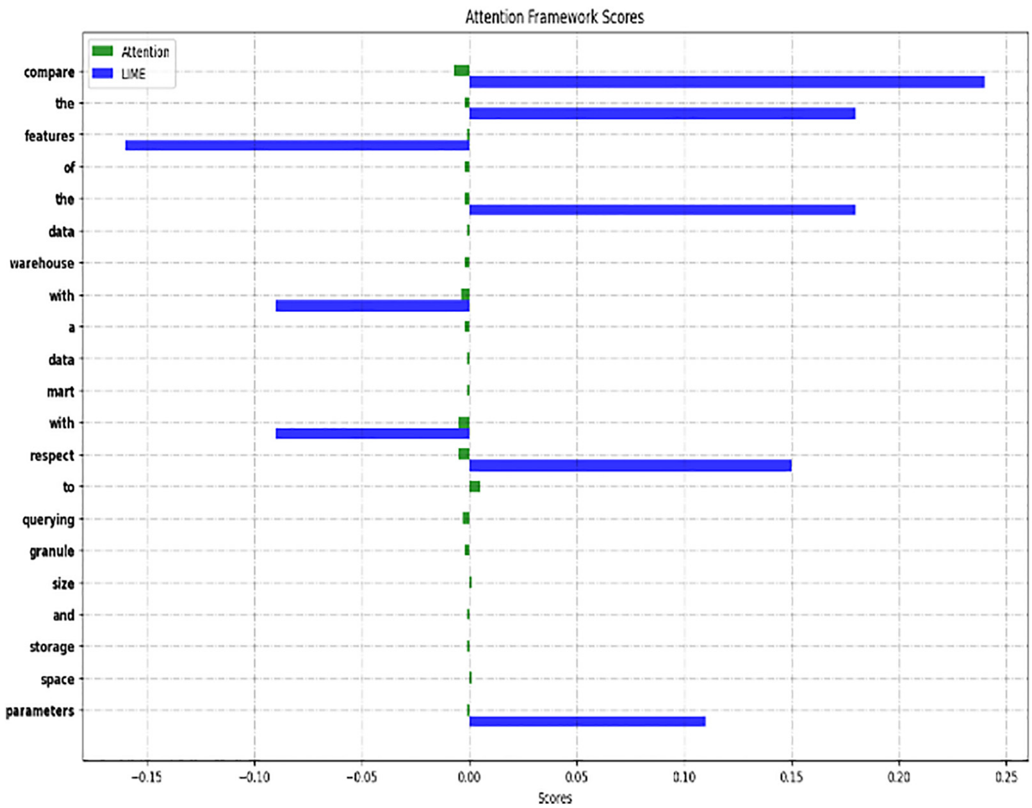

Fig. 10. comparison of score assignment of LIME explainer and attention explainer for the third case.

Effectiveness of our proposed framework of explainability

In this subsection, to answer research question number 3, we assess the performance of our suggested framework's explainability along with the LIME explainer that we modify for our MTL method. Our model is regarded as a "black box."

Fidelity comparison

Consider the following three models: the original model, the model based on the explanations provided by the attention scores, and also the model based on the explanations provided by the LIME explainer. We rely on the “F” fidelity metric to measure the performance of each XAI technique compared to the original model. On the other hand, we compare their performance in addition to our original model with the true label values (Ground truth labels).

Let G, A, and L be three functions representing these three models and T a function representing the true annotation values. We define fidelity as follows:

$$
\begin{aligned}
F_{A,G}(x) &= \begin{cases} 1 & \text{if } A(x) = G(x) \\ 0 & \text{else} \end{cases} \\
F_{L,G}(x) &= \begin{cases} 1 & \text{if } L(x) = G(x) \\ 0 & \text{else} \end{cases} \\
F_{A,T}(x) &= \begin{cases} 1 & \text{if } A(x) = T(x) \\ 0 & \text{else} \end{cases} \\
F_{L,T}(x) &= \begin{cases} 1 & \text{if } L(x) = T(x) \\ 0 & \text{else} \end{cases}
\end{aligned}
\tag{22}
$$

Where x is an item from our learning objectives dataset.

From the results of Table 8, we can deduce that our framework is more faithful to our proposed model and also to the true label values since it reaches a value of 99.08% fidelity against 97.46% for the model based on the LIME explainer.

Computational costs calculation

In addition to the fidelity metric, we take into account the computational cost of each technique. We evaluated the execution time of each method on our entire dataset. For each item in the dataset, our suggested framework needed less than a minute, compared to an average of 20 minutes for each item in the LIME explanation. To get all of the explanations from the

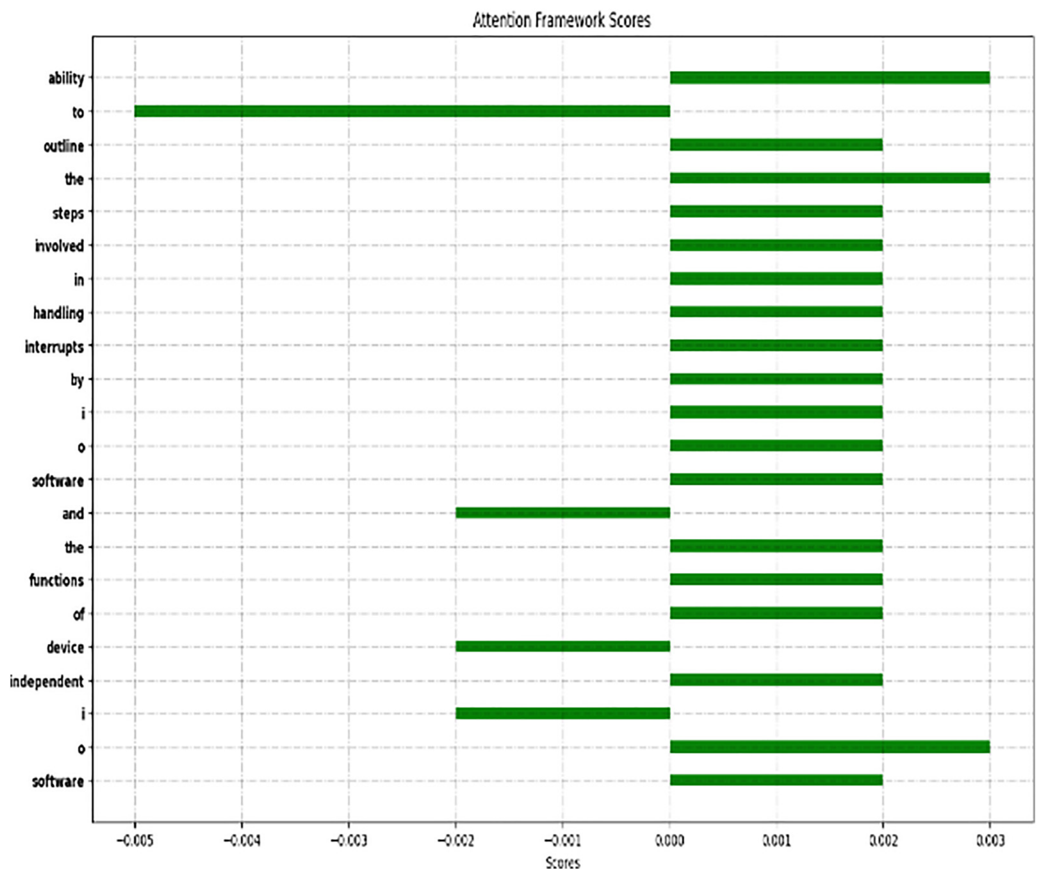

Fig. 11. Attention score assignment of the second case.

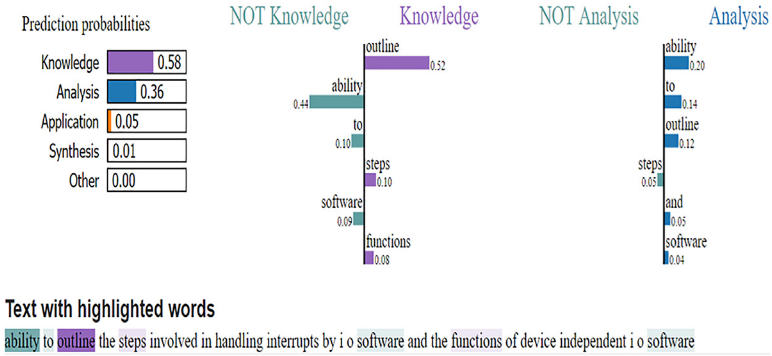

Fig. 12. LIME score assignment of the second case.

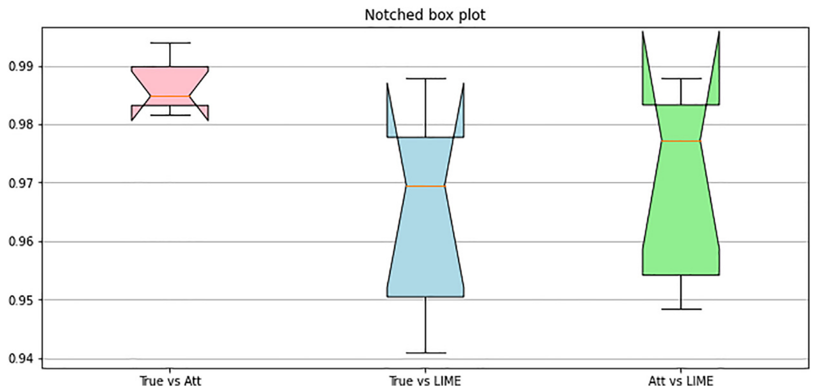

Fig. 13. Box plots for comparing the correlation between the two XAI techniques and the true values.

LIME explainer, we had to enhance the execution using the graphics processing unit (GPU) and split the dataset into many parts. Our entire dataset, which consists of 2400 items, took us more than 20 hours to calculate the explanations because even with the use of GPUs, the execution time for each item is almost half a minute. Because it must recreate the scenario that was previously described in section (4.2.2) for each occurrence and optimize the equation, the LIME explanation is demanding in terms of execution time (15).

Qualitative analysis

In this section, we analyze the performance of our proposed explainability framework through case studies. We study the scores assigned by the two explainability techniques: LIME and our proposed framework. To better appreciate the difference and similarities between the two XAI methods, we show in Table 9 three case studies.

Fig. 9 illustrates the first case presented in Table 9, it shows that when it comes to cases where the two techniques converge to the same prediction, the LIME scores are stronger compared to our framework.

Fig. 10 illustrates the third case presented in Table 9. The third case demonstrates the situations where the two techniques are in error.

Sometimes these XAI methods conflict with each other. In the second case presented in Table 9, the two techniques diverge. as shown in Fig. 11 the attention scores are positive for the two tokens “ability” and “outline”, thereby the attention framework classifies correctly the second case.

On the other hand, LIME classifies the learning objective in the cognitive level of “Knowledge”. Fig. 12 shows that the score assigned to “Knowledge” is 0.58 against 0.36 for “Analysis”. For LIME the token “outline” is representing the knowledge level with a score of 0.52.

To analyze the degree of correlation between the two techniques, we base ourselves on the Pearson coefficient. Fig. 13 shows the box plot of the distribution of correlation coefficients over the whole dataset between the predictions of LIME, our proposed framework as well as the true values. We can notice that these results confirm the results already obtained during the study of the fidelity of the two XAI techniques. The “True vs LIME” and “Att vs LIME” box plots are high, which means that the correlation values are spread due to some diverging values of the LIME explainer. On the other hand, the “True vs Att” box plot is short, which means that the global attention framework predictions are correlated with the true values.

On the other hand, when analyzing the results of our proposed explainability framework, We observed that many word tokens with high attention weights, in some situations, were adjectives or adverbs that explicitly signaled the underlying class name (see Figs. 9–11). The attention scores of words like "to" and "and" in Fig. 12 are negative, which has a negative impact on the explanation and makes the attention mechanism harder to understand. On the other hand, in other situations, we have also noticed that such helpful words might not always be able to receive significant attention weights.

Conclusion

In this article, we present MTBERT-Attention, a unique model based on MTL and the mechanism of co-attention, which simultaneously learns cognitive classification with a relevant secondary task. To the best of our knowledge, no research has been reported that performs cognitive text classification using BERT in conjunction with MTL and the co-attention mechanism. The combination considerably improved the performance of our model given the lack of annotated data based on Bloom's taxonomy's cognitive levels. To validate the performance of our proposed method a variety of baseline models were designed and introduced and a comparative evaluation was conducted.

On the other hand, we have proposed an explainability framework based on the attention mechanism, and we have demonstrated its performance in terms of fidelity and computational cost. However, we noticed that in some cases, tokens that are often adverbs or prepositions receive high scores, which makes the attention mechanism less interpretable. This is due to the global character of explanations versus the local character of LIME explanations.

Ultimately, our suggested method demonstrates its effectiveness for categorizing learning objectives in the context of MOOCs. Our method can be used for various learning objects in the context of E-learning because it is based on learning objectives for the pedagogical classification of MOOCs.

In the future, we will improve the performance of our proposed explainability Framework by adopting and comparing different techniques of attention mechanism.

Data Availability Statement

Data available on request. The data underlying this article will be shared on reasonable request to the corresponding author. *Funding - (Not applicable) *Conflicts of interest/Competing interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. *Ethics approval - (Not applicable) *Consent to participate - (Not applicable) *Consent for publication: I, the author of this paper, give my consent for the publication of identifiable details, which can include photograph(s) and/or videos and/or case history and/or details within the text ("Material") to be published in the above Journal and Article. *Availability of data and material: The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hanane Sebbaq: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation. **Nour-eddine El Faddouli:** Writing – review & editing, Supervision, Validation.

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